Honeywell | Intelligent Life Care



Systevo Electronic Module IP

Part No. 72595F

GB **Installation- and Commissioning Instruction**

89500A6.GB0
08.2024

Intended purpose
The products may only be used for the applications outlined in the catalogue and in the technical documentation, and only in conjunction with the recommended and approved components.
This documentation contains both registered and unregistered trademarks. All trademarks are the property of the respective owners. The use of this documentation does not grant you a licence or any other right to use any names, logos and/or labels mentioned or depicted here.
This documentation is subject to the copyright of Honeywell. The content must not be copied, published, modified, distributed, transmitted, sold or changed without the express prior written permission of Honeywell.
The information contained in this documentation is provided without warranty.

Safety information
This documentation includes all the information required for the proper use of the products described in it.

In order to ensure correct and safe operation of the products, they must be transported, stored, installed and assembled properly and operated with care.
The term 'qualified personnel', in terms of the safety instructions detailed in this documentation or on the product itself, refers to

- project engineers who are familiar with the safety guidelines concerning patient call systems incl. their components.
- trained service engineers who are familiar with the components of patient call systems and the operating information contained in these instructions.
- trained installation and service personnel that have the necessary qualifications for installing / carrying out repairs on patient call systems incl. their components, or who are authorised to operate, ground and label electrical circuits and devices/systems in compliance with the relevant safety standards.

Symbols
The following information is provided in the interest of personal safety and to prevent damage to the products described in these instructions and any connected equipment.
Safety instructions and warnings intended to prevent hazards to the life and health of users and maintenance staff or damage to the equipment are highlighted in the instructions by the symbols detailed below. The symbols used in these instructions have the following meanings:



Warning - Designates risks to person and/or machine. Non-compliance will create risks to person and/or machine. The level of risk is indicated by the word of warning.



Note – Important information about the product or a specific part of the instructions which should be read with particular attention.



Standards and guidelines – Information and requirements as per the national and local guidelines and applicable standards.

Dismantling



In accordance with Directive 2012/19/EU (WEEE), the manufacturer will accept the return of the electrical and electronic device for proper disposal after dismantling.
Detailed information on how to remove batteries safely is available on the website www.ackermann-clino.com.

© Honeywell International Inc. / technical changes reserved!
This documentation is subject to copyright law and, as per Sections 16 and 17 of the German Copyright Act (UrhG), is neither permitted to be copied nor disseminated in any other way. Any infringement as per Section 106 of the UrhG may result in legal action.

Table of contents

1 General / Application ... 4
1.1 Associated documents ... 4
1.2 Abbreviations ... 5
1.3 Requirements ... 6
1.4 Scope of functions ... 6
2 Topologies ... 6
2.1 Possible topologies ... 7
3 Installation ... 15
3.1 Removing the lamp shell / opening the device ... 16
3.2 Removing the circuit board from the bottom housing panel ... 16
3.3 Installation on a wall ... 17
3.4 Mounting on installation socket ... 17
3.5 Cable routing when mounting on a flush installation socket ... 18
3.6 Inserting the connection board into the bottom housing panel ... 18
3.7 Mounting on installation socket (e.g. tunnel socket) ... 19
3.8 Wiring the connecting cables ... 20
3.9 Putting on the cover ... 20
3.10 Mounting the name plate ... 21
3.11 Mounting the name plate as an independent unit ... 22
3.12 Writing on the name plate ... 22
4 Electrical connection ... 23
4.1 Cable types ... 23
4.2 Mains power supply ... 24
4.3 Mode switch ... 25
5 Connecting terminals to the Systevo Electronic Module IP (SEM-IP) ... 26
5.1 Position of the connecting terminals ... 26
5.2 USB-C port ... 27
5.3 Settings of the slide switch for terminating the data lines ... 27
5.4 Connecting terminals and their function ... 28
5.5 System overview ward controller & cluster controller ... 29
5.6 System overview cluster controller ... 30
5.7 System overview cluster controller with SINF connection to the ZBUS ... 31
5.8 IP-connected room (PoE) ... 32
6 Commissioning ... 35
7 Scope of delivery / accessories / licenses ... 36
8 Technical specifications ... 37

Additional and updated information

The features, specifications and product information detailed in this documentation are correct at the time of printing (see cover for date); however, the information specified in this document may differ slightly from the actual product as a result of product changes and/or revised standards and guidelines related to planning, installation and commissioning.

For reference, please visit www.ackermann-clino.com for the latest information updates, compliance statements and servicing specifications.

1 General / Application

The Systevo Electronic Module IP, referred to below as SEM-IP, is the central room controller for IP call systems with and without intercom and can be used in connection with the entire Systevo Call Ackermann range. The pending call status is displayed outside of the patient or duty room with highly efficient multicoloured LED lights (6 lamp chamber). The SEM-IP is available in silver or white and can optionally be equipped with a name plate.
Different operating modes enable a seamless migration and usage in future care communication systems. Owing to the pioneering software and hardware architecture, software updates can be performed during operation without impairing the system's up time. With the aid of the flexible software licences, the Systevo Electronic Module IP can be expanded to support different application cases and requirements.
The components in the patient room are connected with the Systevo Electronic Module IP via the DBUS. The standard-based requirements of VDE 0834/UL 1069 are met, so this device can be utilised in various different markets (EU, USA, Canada, etc.). A standard USB-C slot is available for future applications (WLAN / Bluetooth).

Equipment

- 2 DBUS lines. 4 DBUS units (3+1) are supported per DBUS line.
- 2 I/O ports preconfigured with standard functions to connect passive components.
- Selector switch to set the operating mode.
- Slide switch to configure the termination of the room and audio bus.
- Bottom housing panel with connection board prepared to hold a pluggable light chamber and a name plate
- USB-C standard portal



Safety information:

- In accordance with DIN VDE 0834-1, connecting cables (including cable laying) must be able to withstand a test voltage of 4 KV. The mains power for the devices can be supplied in the same cable. Note the corresponding wire cross-sections!
- Systevo Electronic Modules are guaranteed to work properly only if mounted and installed correctly in accordance with Fig. 1–5.
- A functional test is mandatory to ensure safe operation in accordance with current regulations (DIN VDE 0834-1:2016-06) (see Chapter entitled "Functional test").

1.1 Associated documents

Description	Part No.
Installation Instruction Systevo Bed Units	89470K1.GB0
Installation Instruction Systevo Care View IP	89470K3
Installation Instruction Systevo Audio Interface & Audio Switch SAI & SAS	89475A2
Installation Instruction Systevo Room Terminal SRT	89475A3
Installation Instruction Systevo Passive Unit SPU	89475A4
Installation Instruction Systevo System	89470A5.GB0 / A6.GB0
Commissioning of the Clino System 99plus Patient Call System	89440C7

1.2 Abbreviations

ABUS	:	Audio bus
IP-PABX	:	Software-based telephone system
CoS	.	Class of service (different services are classified via CoS)
CP99Config	:	Configuration program for Clino Systevo
DBUS	:	Data bus
DTMF	:	Dual-tone multi-frequency
ESPA	:	Enhanced signalling protocol for alarm processes
Intercom	:	Call answering, responding to patients, calls to IP-enabled devices
IP	:	Internet Protocol
LAN	:	Local Area Network
MAC	:	Media Access Control
MMS	:	Management server, provides standardised network services and manages the Systevo Touch IP terminals and SEM IPs
Multicast	:	Broadcasting of packages from one sender to a group of recipients
PABX	:	Private Automatic Branch eXchange (telephone system)
PoE	:	Power over Ethernet
PSA	:	Personal search device
RABUS	:	Room audio bus
RFC	:	System request for comments (technical documentation)
Routing	:	Controls the routes of IP packages
RTP	:	Real-time transport protocol
SEM+	:	Systevo Electronic Module + Audio
SEM-IP		Systevo Electronic Module IP
SINF		Systevo Information Display
SIP	:	Session Initialisation Protocol
SCL	:	Systevo Corridor Lamp (room signal lamp)
SSID	:	Service Set Identifier, name of a WLAN network
STIP	:	Systevo Touch Terminal IP
SCVIP	:	Systevo Care View IP
SCBIP	:	Systevo Connection Board (connection board for STIP / SCVIP)
STP	:	Spanning Tree Protocol
TK	:	Telecommunications system
TTL	:	Time to live (validity period of the data in networks)
VoEth	:	Voice over Ethernet (proprietary)
VoIP	:	Voice over IP
VM	:	Virtual machine
WiFi	:	Wireless Fidelity
WLAN	:	Wireless Local Area Network
WM-LAN	:	Wall module with local network connection
ZBUS	:	Room bus

1.3 Requirements

The Systevo Electronic Module IP (SEM-IP) is operated in system environments equal to or higher than V12.14. Licences are needed depending on the requirements as a cluster controller / ward controller. No licences are required for the IP-connected room function.

1.4 Scope of functions

In the delivery state (mode switch position 1), the SEM-IP functions as an IP-connected room controller. It is also possible to use the SEM-IP as a ward / cluster controller (see also Chapter 4.3).



Please note: The full range of functions will only be available at a later date.

2 Topologies

The Systevo Electronic Module IP (SEM-IP) can be used in a wide range of topologies depending on the requirements. As a “cluster controller”, it is connected to the nurse call system via the LAN portal and in this function as a ward controller is able to supply a room as a room controller and up to 126 (licences required) additional devices on the ZBUS. In this function, the SEM-IP can be viewed as a ward controller and works equally well in conjunction with Systevo control panels SMCc and ward controllers of the SCIP+ type. A flexible adjustment of the rooms to be connected is possible with various licences.

2.1 Possible topologies

2.1.1 SEM-IP as a ward / cluster controller in mixed operation with SMCc / SCIP+

The SEM-IP is preset to function as a room controller with the mode switch position 1 as standard on delivery. The 79SW420 licence is required for the cluster controller function (mode switch position 2/3). As a cluster controller, the SEM-IP takes on the function of a ward controller, comparable to SMCc / SCIP+. Mixed operation of the various controller versions is possible. The prerequisite for this is the use of the same firmware versions V12.14 or newer in all controllers.

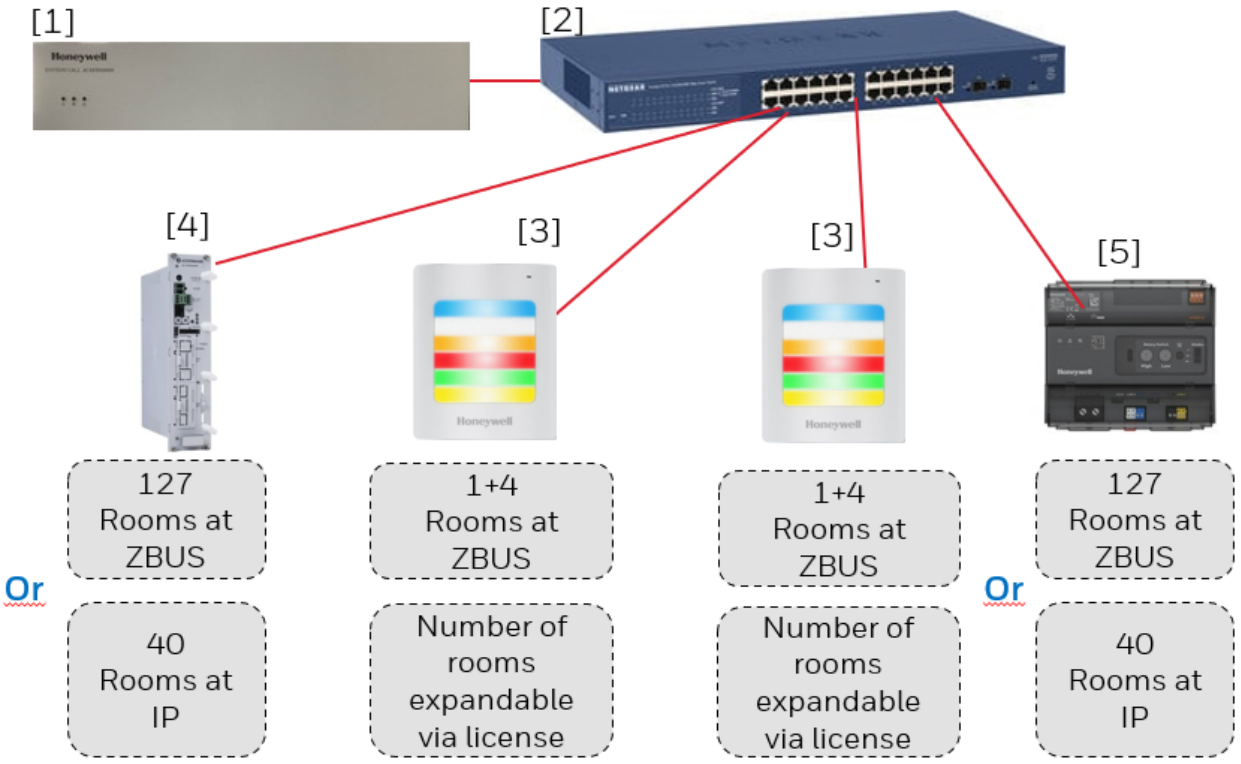


Fig. 1: System overview ward controller & cluster controller

[1]	Systevo server
[2]	System switch
[3]	Cluster controller SEM-IP
[4]	Ward controller SMCc
[5]	Systevo control IP+ SCIP+

Controller type	Max. number of ZBUS devices (rooms)	Max. number of IP devices (rooms)
SMCc	127	40
SCIP+	127	40
SEM-IP	1+126 (licences required)	40

Licences:

Part No.	Name	Scope of functions	Note
79SW420	SEM-IP as cluster controller	1 + 4 rooms on the ZBUS.	1 licence each for SEM-IP with mode switch position 2 or 3
79SW421	Extension licence 5 rooms	Counts **ALL** rooms in the entire system under the cluster controllers that are not covered by 79SW420.	Applies system-wide
79SW422	Extension licence 20 rooms	Counts **ALL** rooms in the entire system under the cluster controllers that are not covered by 79SW420.	Applies system-wide

- The mode switch position 2 & 3 and the corresponding functions are identical to the mode 2 & 3 functions of the control panel (SCIP+).
- If SEM-IP is used in mix operation with SMCc or SCIP+, all controllers must be equipped with firmware V12.14.

2.1.2 SEM-IP as a cluster controller (1+4 rooms)

As a cluster controller, the SEM-IP is connected to the system via the LAN portal and also enables the connection of room components to its own DBUS. In this function, up to 126 additional ZBUS devices can be connected to the SEM-IP. (Licences required) A separately laid tone wire to the central 24 V DC power supply unit is used for the mains power.

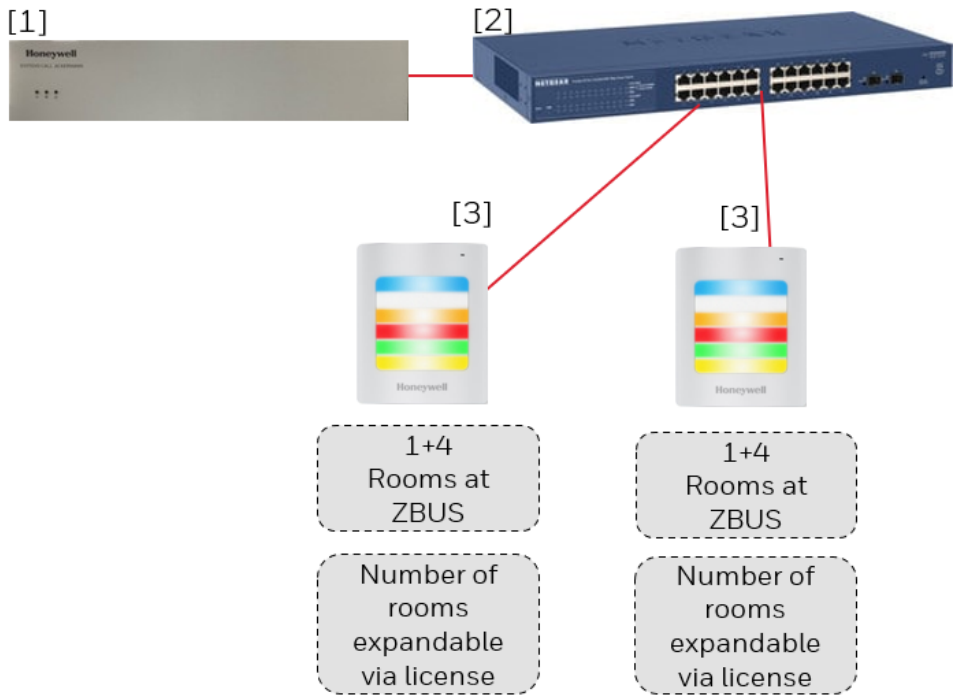


Fig. 2: System overview cluster controller

[1]	Systevo server
[2]	System switch
[3]	Cluster controller SEM-IP
[4]	Ward controller SMCc
[5]	Systevo control IP+ SCIP+

Licences:

Part No.	Name	Scope of functions	Note
79SW420	SEM-IP as cluster controller	1+4 rooms on the ZBUS.	1 licence each for SEM-IP with mode switch position 2 or 3
79SW421	Extension licence 5 rooms	Counts **ALL** rooms in the entire system under the cluster controllers that are not covered by 79SW420.	Applies system-wide
79SW422	Extension licence 20 rooms	Counts **ALL** rooms in the entire system under the cluster controllers that are not covered by 79SW420.	Applies system-wide

Simplistic mains power with central 24 V DC power supply unit. PoE cannot be used in this function mode.

2.1.3 SEM-IP as a cluster controller (1+4 rooms) with SINF connection to the ZBUS

As described in Chapter 2.2. Also with the option to connect the Systevo Information Display (SINF) to the ZBUS, in which case one extension license 79SW419 is required for each single/double display.

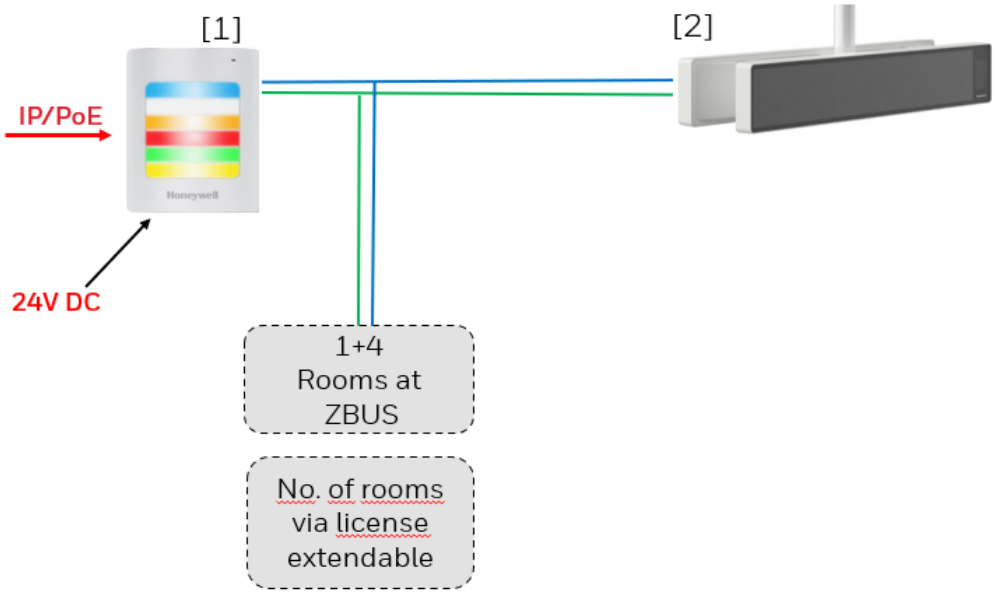


Fig. 3: System overview cluster controller (1+4 rooms) with SINF connection to the ZBUS

[1]	Cluster controller SEM-IP
[2]	Systevo Information Display (SINF)

Licences:

Part No.	Name	Scope of functions	Note
79SW419	Extension license for the Systevo Information Display	Connection of an SINF to the DBUS / ZBUS of the SEM-IP	**ZBUS:** 1 license for each single/double SINF **DBUS:** 1 license for each SINF
79SW420	SEM-IP as cluster controller	1 + 4 rooms possible with DBUS	1 licence each for SEM-IP with mode switch position 2 or 3.
79SW421	Extension licence 5 rooms	Counts **ALL** rooms in the entire system under the cluster controllers	Applies system-wide
79SW422	Extension licence 20 rooms	Counts **ALL** rooms in the entire system under the cluster controllers	Applies system-wide

- Simplistic mains power with central 24 V DC power supply unit. PoE cannot be used in this function mode.
- **One** additional license 79SW419 is required for each Systevo Information Display (single / double).
- The number of rooms can be expanded using a license.
- Limitation of a maximum of 6 single / double displays on one SEM-IP.

2.1.4 SEM-IP as an IP-connected room controller

With the mode switch position 1, the SEM-IP works as a room controller with a connection to the nurse call via IP and thus enables the creation and operation of complex wards in IP-only operation. No additional licenses are required for this operating mode.

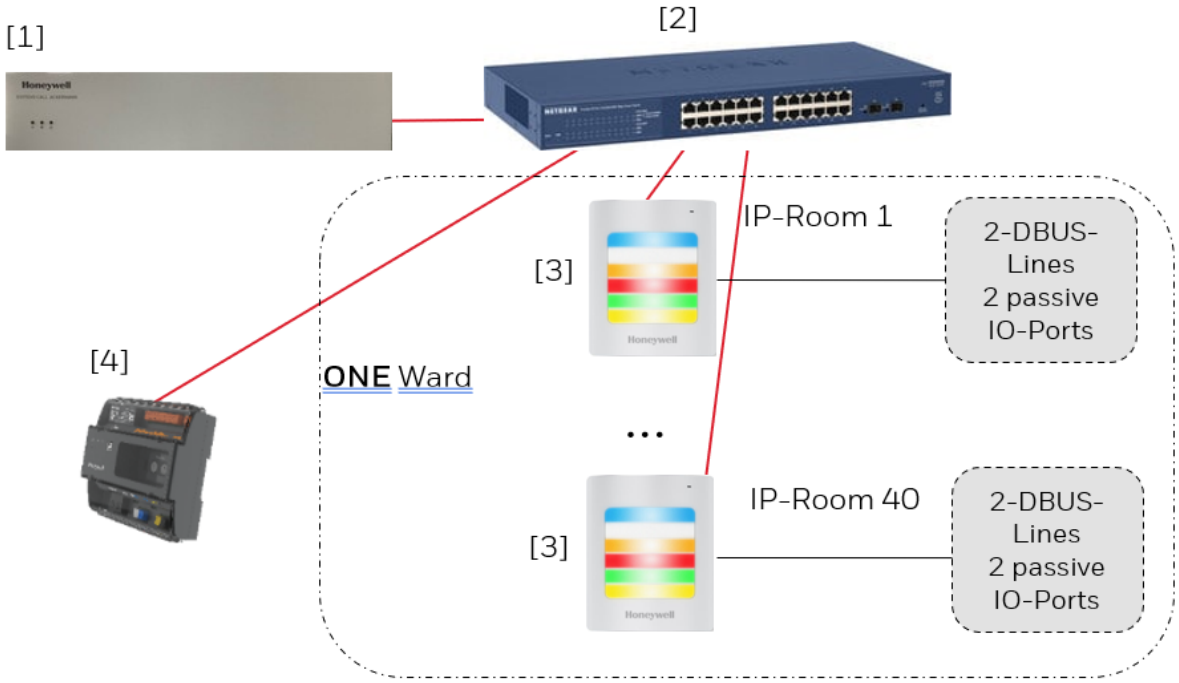


Fig. 4: SEM-IP as an IP-connected room controller

[1]	Systevo server
[2]	System switch
[3]	Systevo Electronic Module IP
[4]	Systevo Control SCIP

- 64 ward controllers [4] are possible in total.
- 40 SEM-IP per ward (logical unit).
- 2560 IP-connected rooms in total.

2.1.5 SEM-IP as an IP-connected duty room with option as an IP gateway for operating SINF

The SEM-IP is connected to the nurse call as an IP device either via LAN or PoE and enables the direct connection of Systevo Information Displays to the DBUS. An external mains power supply on the SEM-IP provides the mains power for the Systevo Information Display SINF. The Systevo Care View IP is conveniently connected via PoE and assigned to the associated SEM-IP via a logical connection. The Systevo Care Unit SCU ensures **standard-compliant function** if the Systevo Care View IP is unplugged!

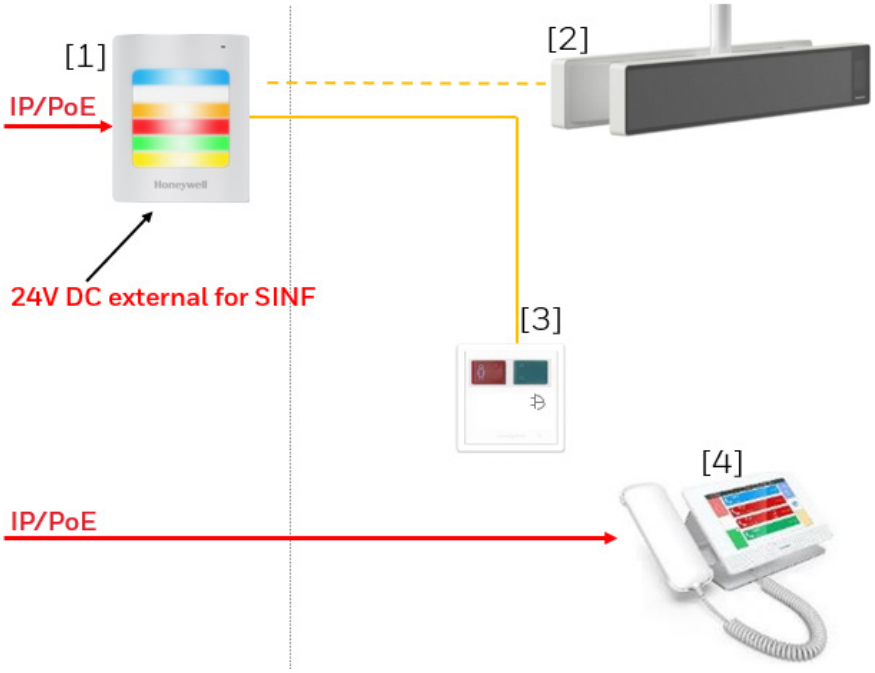


Fig. 5: SEM-IP as an IP-connected duty room with IP gateway for operating SINF

[1]	Systevo Electronic Module IP SEM-IP
[2]	Systevo Information Display SINF
[3]	Systevo Care Unit SCU
[4]	Systevo Care View IP SCVIP

- An additional 24 V DC power supply is used for the SINF.
- Double-sided SINFs are managed as two single SINFs in the system.
- Every SINF is connected to the DBUS, mode switch position 2 & 3
- A maximum of 2 SINFs (individual) can be connected.
- The Systevo Care Unit SCU ensures **standard-compliant function** if the Systevo Care View IP is unplugged!
- The logical grouping of Systevo Electronic Module IP (SEM-IP) and Systevo Care View IP (SCVIP) to form a duty room is carried out in the configuration program and is referred to as logical room grouping.

Licences:

Part No.	Name	Scope of functions	Note
79SW419	Extension license for the Systevo Information Display	Connection of an SINF to the DBUS / ZBUS of the SEM-IP	**ZBUS:** 1 license for each single/double SINF **DBUS:** 1 license for each SINF

2.1.6 SEM-IP as an IP gateway for operating SINF

The SEM-IP can be used as an IP gateway, which establishes the data connection of the SINF to the system in IP-only systems.



Fig. 6: SEM-IP as a gateway for the SINF data connection

[1]	Systevo Electronic Module IP SEM-IP
[2]	Systevo Information Display SINF

- No license is required for the SEM-IP function as an IP gateway without connecting additional room components.
- An additional 24 V DC power supply is used for the SINF.
- Double-sided SINFs are managed as two single SINFs in the system.
- Every SINF is connected to the DBUS, mode switch position 2 & 3
- A maximum of 2 SINFs (individual) can be connected
- A dummy cover 72595L0 is available for the SEM-IP

2.1.7 SEM-IP as an IP-connected room controller with SINF connection to the DBUS

The SEM-IP as a room controller can also be used as an IP gateway for the data connection of the SINF to the system.

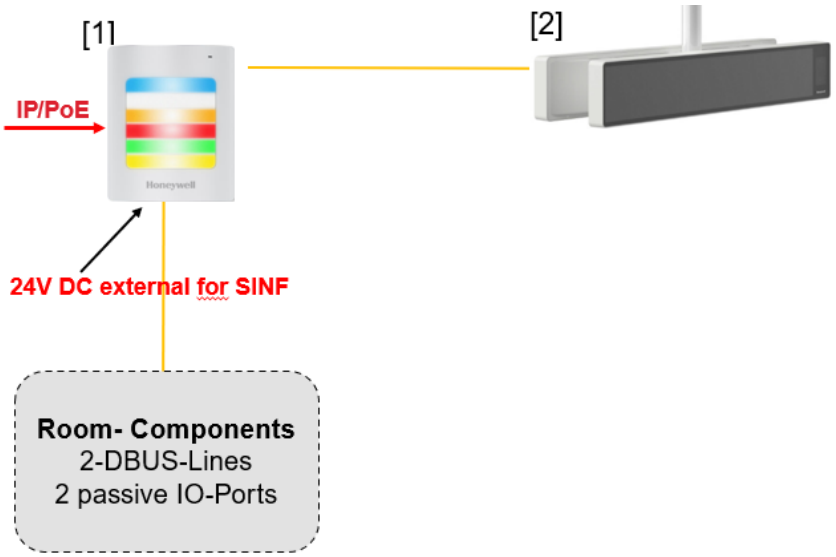


Fig. 7: SEM-IP as an IP-connected room controller and as a gateway for the data connection of SINF

[1]	Systevo Electronic Module IP SEM-IP
[2]	Systevo Information Display SINF

Licences:

Part No.	Name	Scope of functions	Note
79SW419	Extension license for the Systevo Information Display	Connection of an SINF to the DBUS of the SEM-IP	**DBUS:** 1 license for each SINF

- For the SEM-IP function as a room controller, one license 79SW419 is required for each SINF.
- The number of SINFs on the DBUS reduces the number of possible room components on the DBUS.
- An additional 24 V DC power supply is used for the SINF.
- Double-sided SINFs are managed as two single SINFs in the system.
- Every SINF is connected to the DBUS, mode switch position 2 & 3
- A maximum of 2 SINFs (individual) can be connected.

3 Installation

The connecting terminals are suitable for installation cables with a wire diameter of 0.6 mm / 0.8 mm.

Dimensions

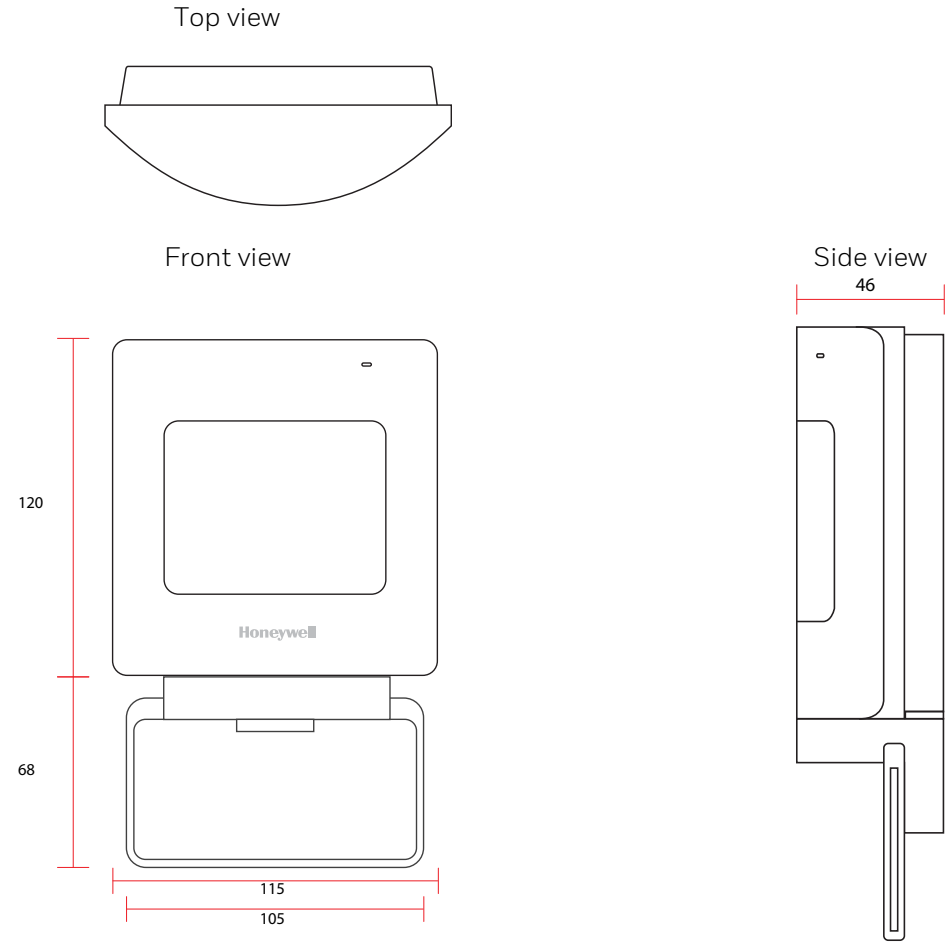


Fig. 8: Dimensions of the Systevo Electronic Module SEM IP with name plate

All mounting and installation work must be carried out only when the device is disconnected from the power.

3.1 Removing the lamp shell / opening the device

The lamp cover on the front side is connected to the bottom housing panel via catches without screw connections and can be easily pulled out without a tool.

3.2 Removing the circuit board from the bottom housing panel

To install the unit on an installation socket or on the wall, the circuit board must first be removed from the bottom housing panel. To do this, press the circuit board out of both catches on the right side and remove it from the guide hooks on the left side.

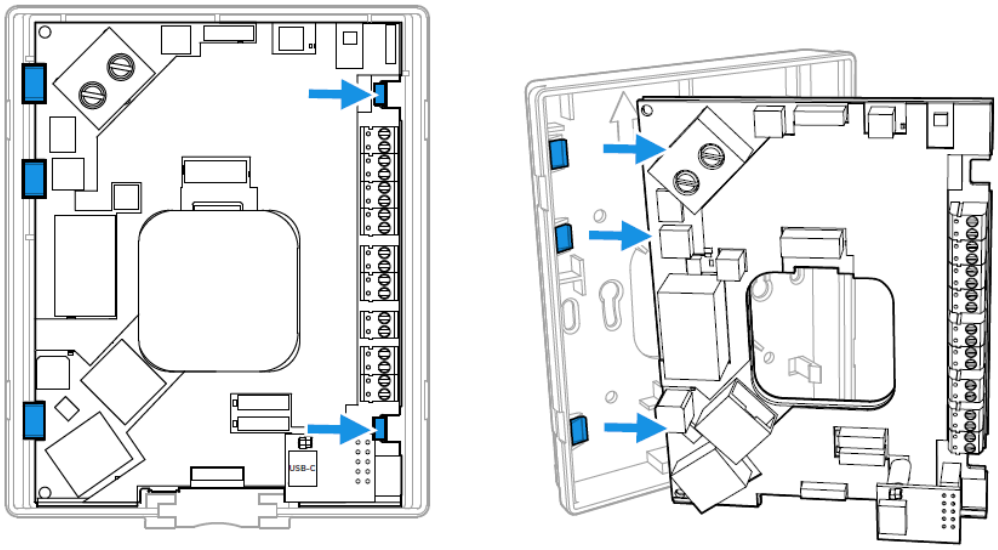
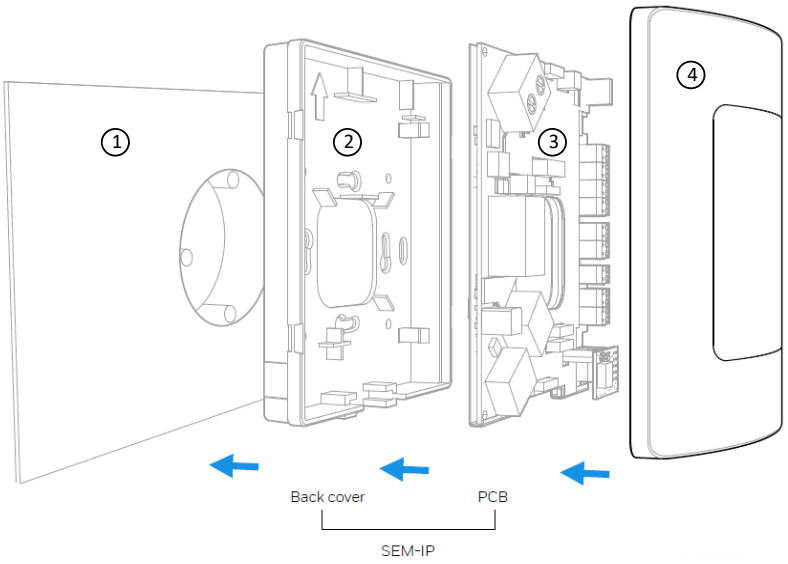


Fig. 9: Removing the circuit board from the bottom housing panel

3.3 Installation on a wall



①	Wall
②	Bottom housing panel
③	Circuit board
④	Cover with lamp shell

Fig. 10: Mounting on the wall

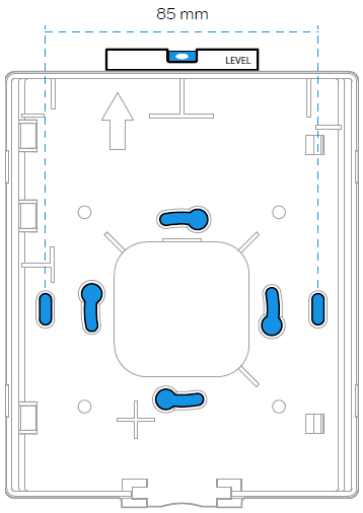
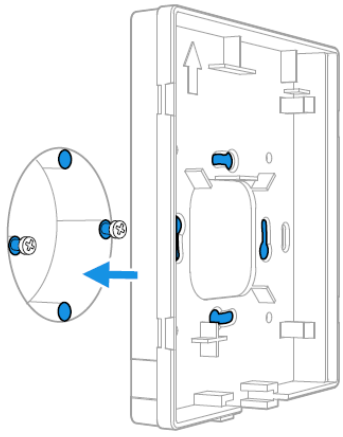


Fig. 11: Mark the external fastening points

- To fasten on a wall (aP), hold the bottom housing panel horizontally (align using a spirit level) against the mounting surface and mark the two external fastening points.
- Drill through the marks to create two holes and insert wall plugs measuring 5 / 6 mm.
- Fasten the bottom housing panel onto the mounting surface using suitable screws.
- The suitable type of wall plugs and screws depends on the material of the mounting surface (cavity wall, concrete, wood etc.) and must be chosen accordingly.

3.4 Mounting on installation socket



Place the bottom housing panel on the installation socket in the correct position, align it horizontally and screw it on using both installation socket screws.

Fig. 12: Fastening on a standard installation socket

3.5 Cable routing when mounting on a flush installation socket

The connections on the circuit board for the ward cabling and the room cabling are clearly separated from each other. Pre-made labels are included in the packaging for easy identification of the installation cables.

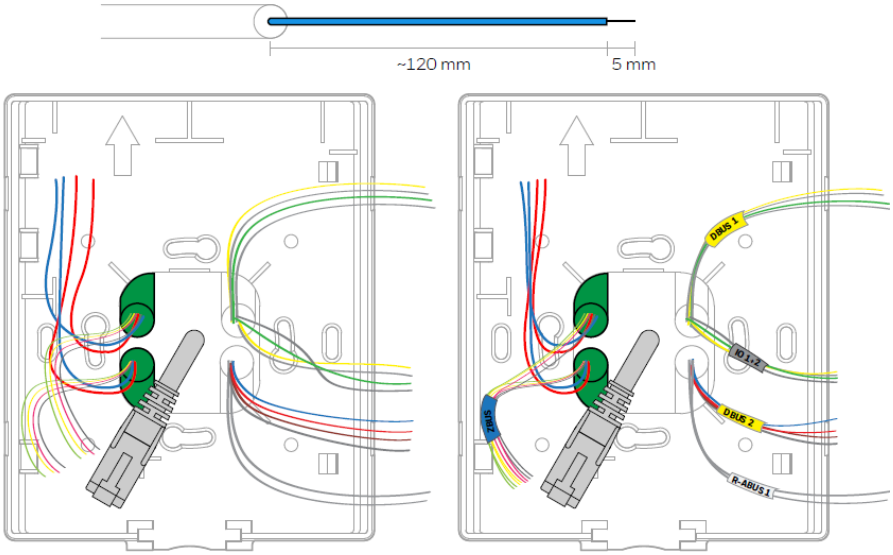


Fig. 13: Cable routing when mounting on a standard installation socket

1. Strip the installation cables to approx. 120 mm and strip individual wires by 5 mm
2. Label cables using the pre-made labels
3. Route the ward cables (e.g. system cables) on the left side and the cables for the room cabling (DBUS, ABUS, IO) on the right side

We recommend using cable material with a maximum wire thickness of 0.6 mm for the room cabling. Thicker wires are much more difficult to accommodate in the housing.

3.6 Inserting the connection board into the bottom housing panel

Push the circuit board onto the guide hooks on the left side.

Then, press the circuit board onto the catches on the right side.

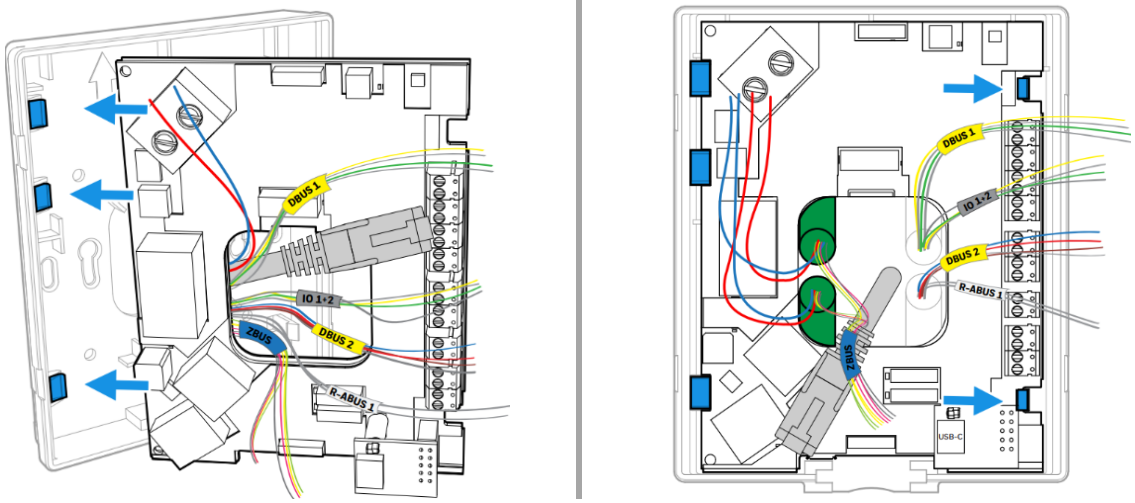


Fig. 14: Inserting the circuit board into the bottom housing panel

3.7 Mounting on installation socket (e.g. tunnel socket)

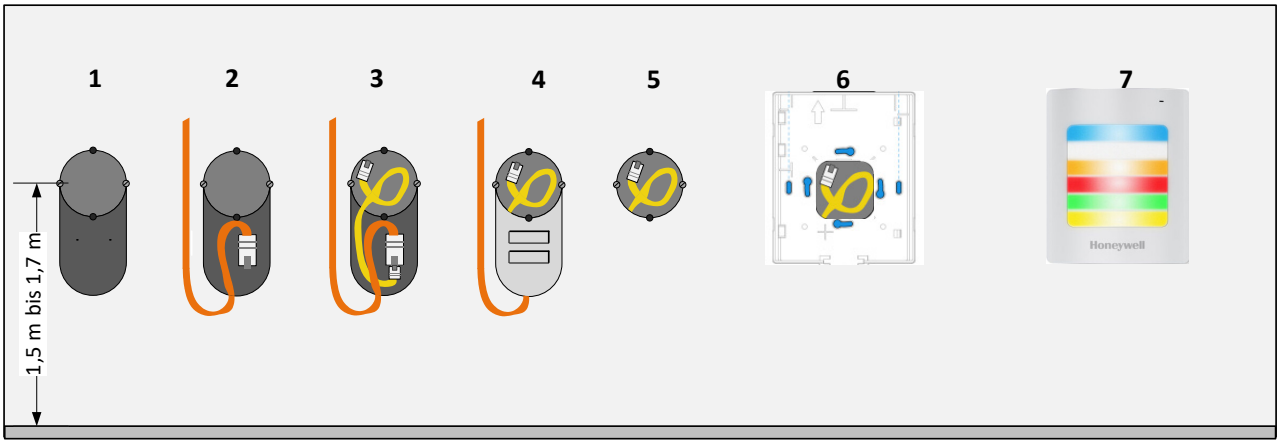


Fig. 15: Keep the mounting height 1.5 m to 1.7 m over the operator's standing surface in accordance with DIN VDE 0834

Installation steps	
1	Flush tunnel box embedded in the wall.
2	CAT cable inserted and data connection equipment with RJ45 Ethernet socket.
3	Use the patch cable included with the connection board.
4	View with closed tunnel.
5	CAT cable and tunnel concealed.
6	Installation of the connection board on the tunnel socket.
7	Connect and install the SEM IP.

In practice, the "tunnel" of the installation socket is plastered over and thus not visible.

3.8 Wiring the connecting cables

Attach the connection terminals and route the cables as shown in the figure.

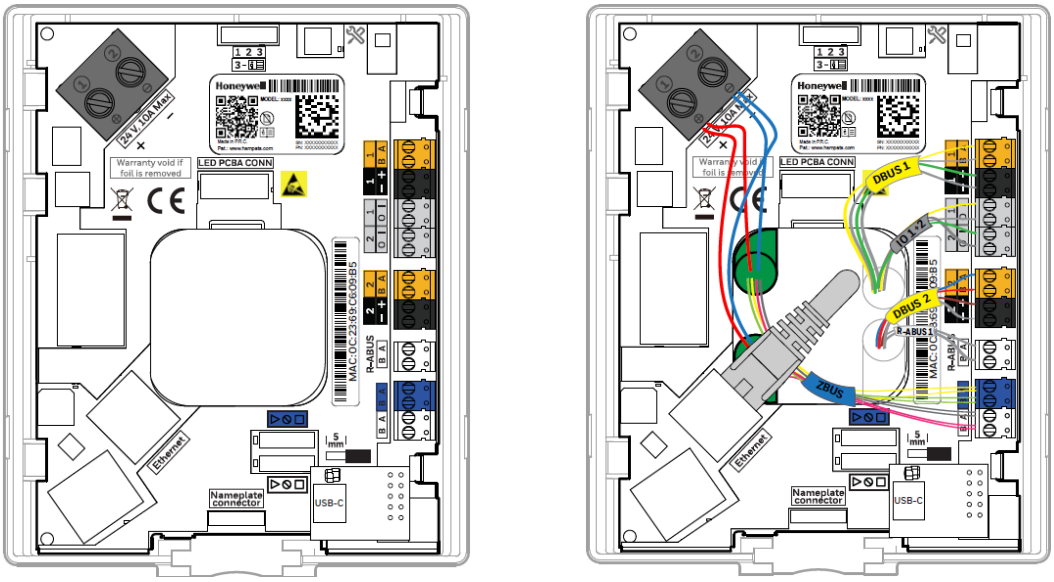


Fig. 16: Attach the connecting terminals and route the cables

3.9 Putting on the cover

- Connect the ribbon cable of the LED board to the main board.
- Place the cover with the lamp shell in the correct position and connect it to the bottom housing panel simply by pressing.

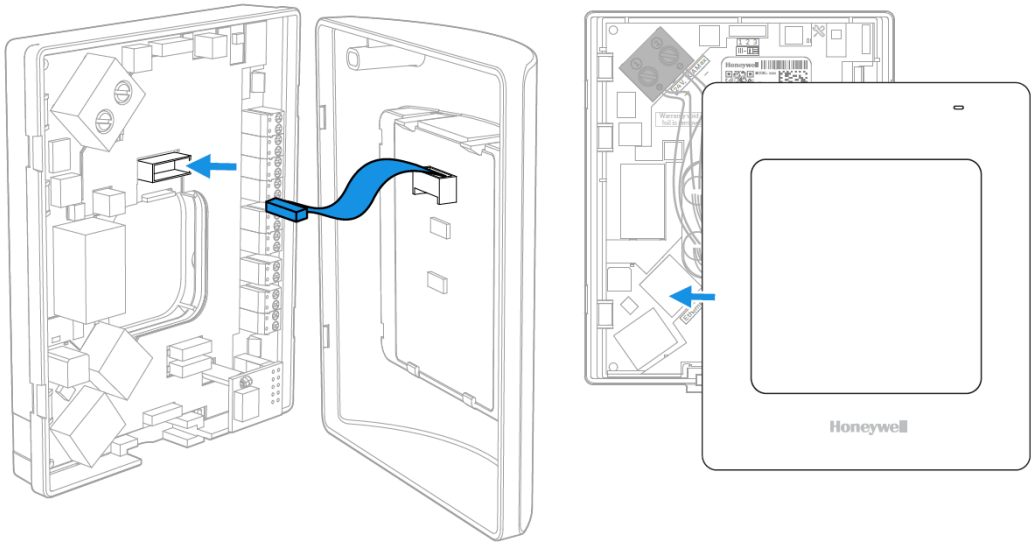
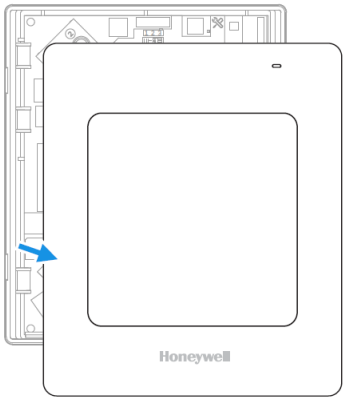
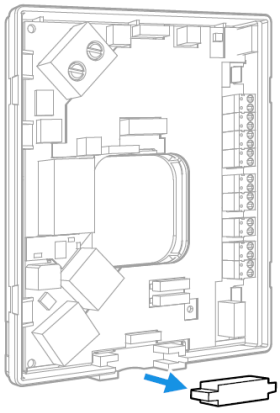


Fig. 17: Put on the cover with the lamp shell

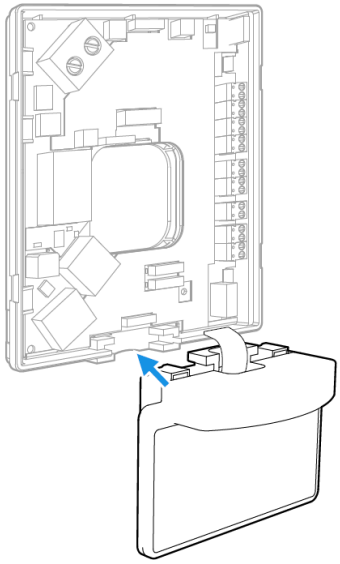
3.10 Mounting the name plate



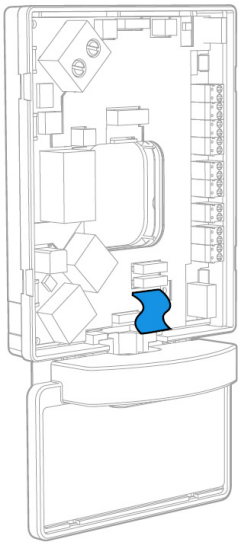
1. Remove the cover with the lamp shell from the bottom housing panel.



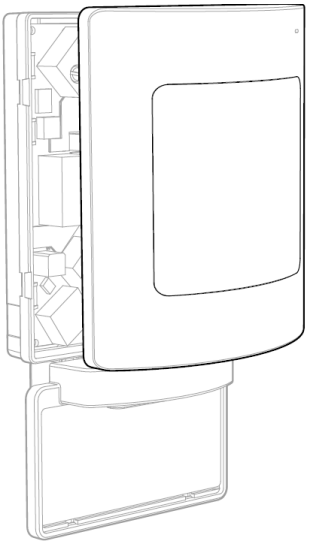
2. Uncover the opening for the connection cable by removing the rubber closure.



3. Slide the name plate into the guide on the underside bottom housing panel.

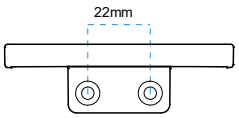


4. Attach the ribbon cable to the name plate connector. The blue marking on the connecting cable must match the blue marking on the protective film.

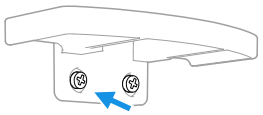


5. Position the cover with the lamp shell and connect it to the bottom housing panel simply by pressing.

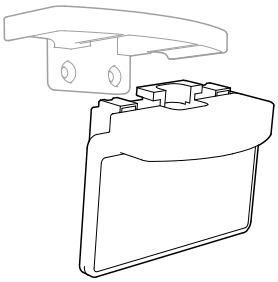
3.11 Mounting the name plate as an independent unit



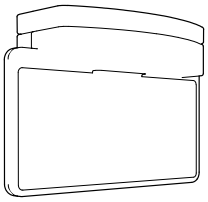
1. Mark the drill holes using the wall bracket as an aid, drill holes corresponding to the wall plugs (5 / 6 mm) and use suitable wall plugs.



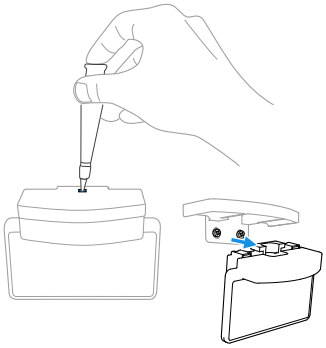
2. Attach the wall bracket to the wall.



3. Slide the name tag onto the wall bracket.



Mounted name plate

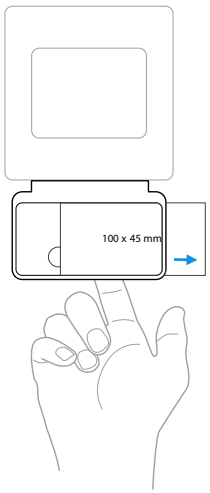


Dismantling

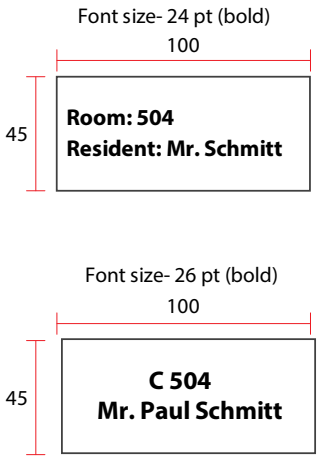
With a tool in the slot, press lightly to release the catch and pull the name plate off the holder.

If you do not release the catch, the mechanism may be damaged upon removing the name plate.

3.12 Writing on the name plate



There is an opening on the back that allows the paper to be pushed out of the name plate to the side.



Recommended font sizes for the paper for labelling.

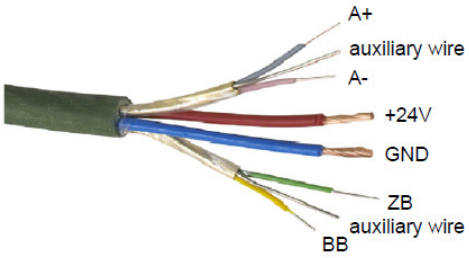
4 Electrical connection

4.1 Cable types

The following table shows available cable types that are recommended and can be used to install the call system components. Different cable types must meet the same technical specifications.

Part No.	Description
89734AH	System cable, halogen-free (drum 500 m) for power supply, time multiplex and voice transmission, 6-wire, with a proof voltage of 2 KV.
89734AJ	System cable, halogen-free (roll 100 m) for power supply, time multiplex and voice transmission, 6-wire, with a proof voltage of 2 KV.
89734AK	System cable, halogen-free (drum 300 m) for power supply, time multiplex and voice transmission, 6-wire, with a proof voltage of 2 KV.

The system cable (Part No. 89734AH/AJ/AK) is recommended for connection to the call system components and for the mains power supply.



The wire must be cut, insulated and arranged so that there are no electrically conductive connections to other wires.

Fig. 18: System cable

- Alternatively, similar 4 x 2 x 0.8 mm cable types can be used for connecting the components for the audio and room data bus. The wire cross-section for the mains power supply must be 2 x 1.5 mm².
- The connecting cables used and the arrangement must be designed in such a way that they withstand a test voltage of 4 KV in compliance with DIN VDE 0834-1.

4.2 Mains power supply

The SEM-IP can optionally be supplied directly through the network connection via PoE. The components in the room can also be supplied this way. An output of 6.75 W **(corresponds to 24 V DC / 280 mA)** is available to supply the components in the room. The maximum output requirement must be observed and not exceeded (see the table below).
If the SEM-IP is used as a ward cluster controller, the mains power is provided via a 24 V DC +/- 10% supply. This also applies if the SINF is connected via the SEM-IP.

Output requirement

Components		Current consumption when idle (mA)	Current consumption max. (mA)	Note Current consumption max.
Systevo Care Unit	SCU	8	11	---
Systevo Display Unit	SDU	8	26	---
Systevo Wall Unit	SBU-DM2+ / SBU-D2+	8	15	---
Systevo Wall Unit	SBU-M2/M5	11	15	---
Systevo Call Easy	SCE	10	30	---
Systevo Call Smart	SCS	62	85	Call + audio + display brightness max.
Systevo Audio Switch	SAS	3	54	Only during switching operation
Systevo Audio Interface	SAI	21	28	---
Systevo call unit with call button	PU	1	8	LED on
Systevo cancel button	PU	1	17	LED on + buzzer (buzzer not from Systevo)
Systevo call / cancel button	PU	1	33	All LEDs on

4.3 Mode switch

The mode switch is used to specify whether the SEM-IP is working as a room controller or as a cluster controller. If the mode switch is in position 1 (factory setting IP address via DHCP server), the SEM-IP will take on the function of a room controller.
As a cluster controller (mode switch in position 2 or 3), the SEM-IP works as a central control and management unit for a logical area (ward). In position 2, the IP address is transferred via a DHCP server and in position 3, a fixed IP address is used. Furthermore, the SEM-IP takes on the function of a room controller in this operating mode and provides the LON bus via the data bus (ZBUS / ABUS) to connect up to 126 additional devices on the data bus. Up to 4 devices are included, and all others are assigned to the cluster (logical area / ward) via licenses. The connection to the system is made via the LAN network interface. If a switch is used with PoE, the power supply is provided via this switch. Alternatively, the power can be supplied via an external 24 V DC power supply.

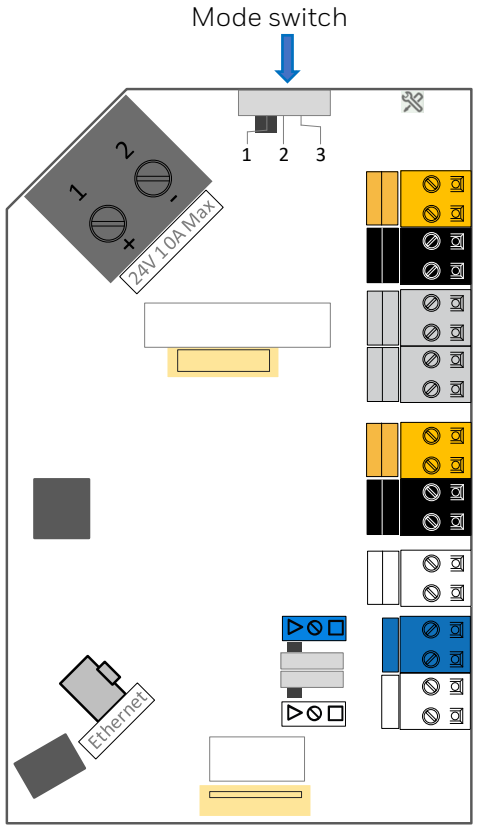


Fig. 19: Mode switch

Mode switch position	SEM-IP mode of operation	Availability
I	IP room controller (IP – address via DHCP)	Q4 / 2024
II	Cluster controller 1+ 4 (IP – address via DHCP)	Available upon request
III	Cluster controller 1+ 4 (permanent IP – address)	Available upon request

The mode switch position 2 & 3 and the corresponding functions are identical to the mode 2 & 3 functions of the control panel (SCIP+).

5 Connecting terminals to the Systevo Electronic Module IP (SEM-IP)

5.1 Position of the connecting terminals

Protective film	Position of the connecting terminals

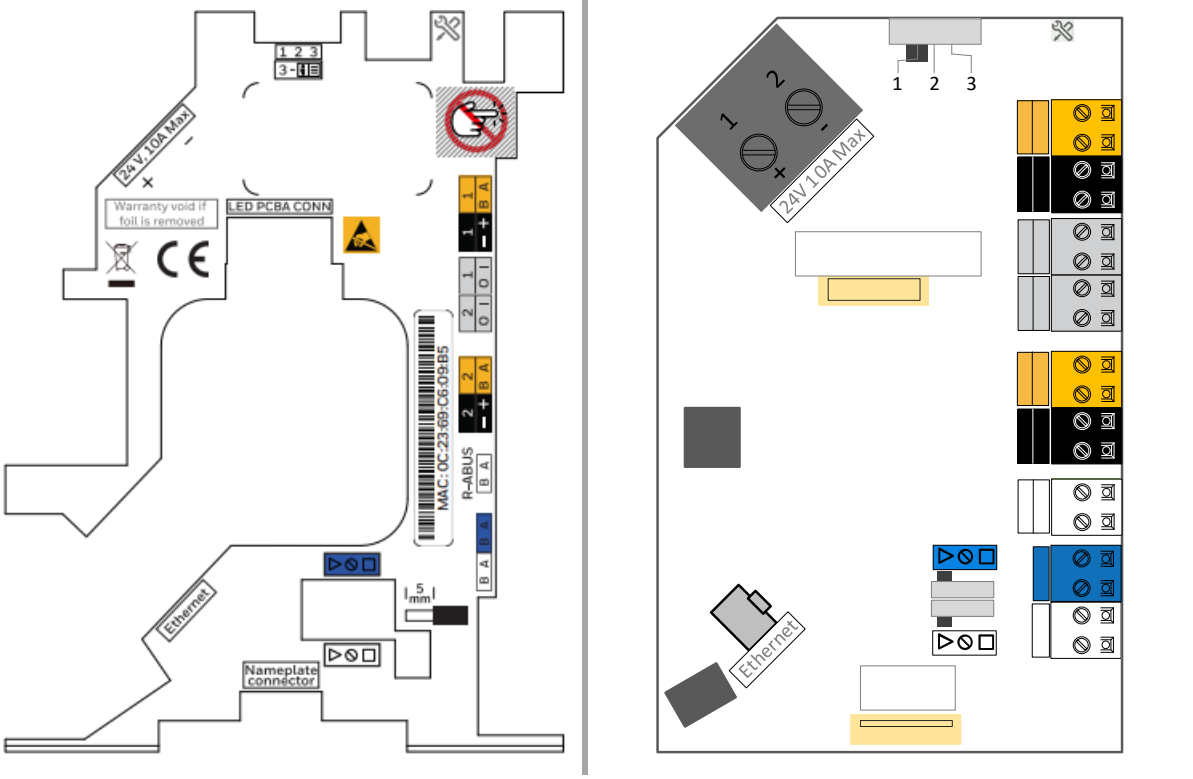
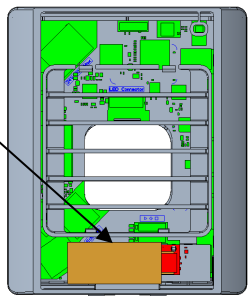
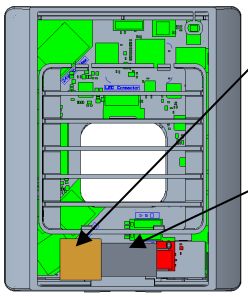
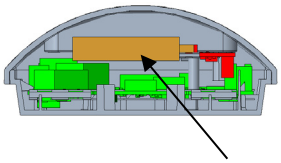
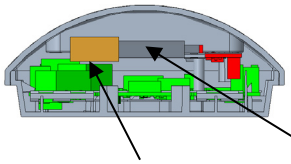


Fig. 20: Protective film and position of the connecting terminals

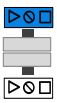
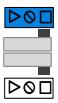
5.2 USB-C port

A USB-C slot is available for future applications (WLAN/Bluetooth).

For dongle with USB-C	For dongle with USB-A
USB dongle L x W x H: 45 x 20 x 10 mm / +/- 2 mm 	 USB dongle L x W x H: 20 x 20 x 10 mm / +/- 2 mm USB adapter C to A L x W x H: 26 x 16 x 7 mm
 Location and position of the USB dongle	 Location and position of the USB dongle and the USB adapter

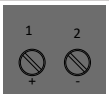
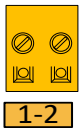
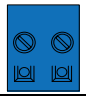
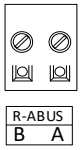
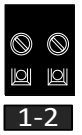
5.3 Settings of the slide switch for terminating the data lines

The slide switches for the ABUS and ZBUS are only effective when functioning as a cluster controller (mode switch position 2 or 3) and enable termination of the respective data line. The termination is enabled as the bus start in the factory setting.

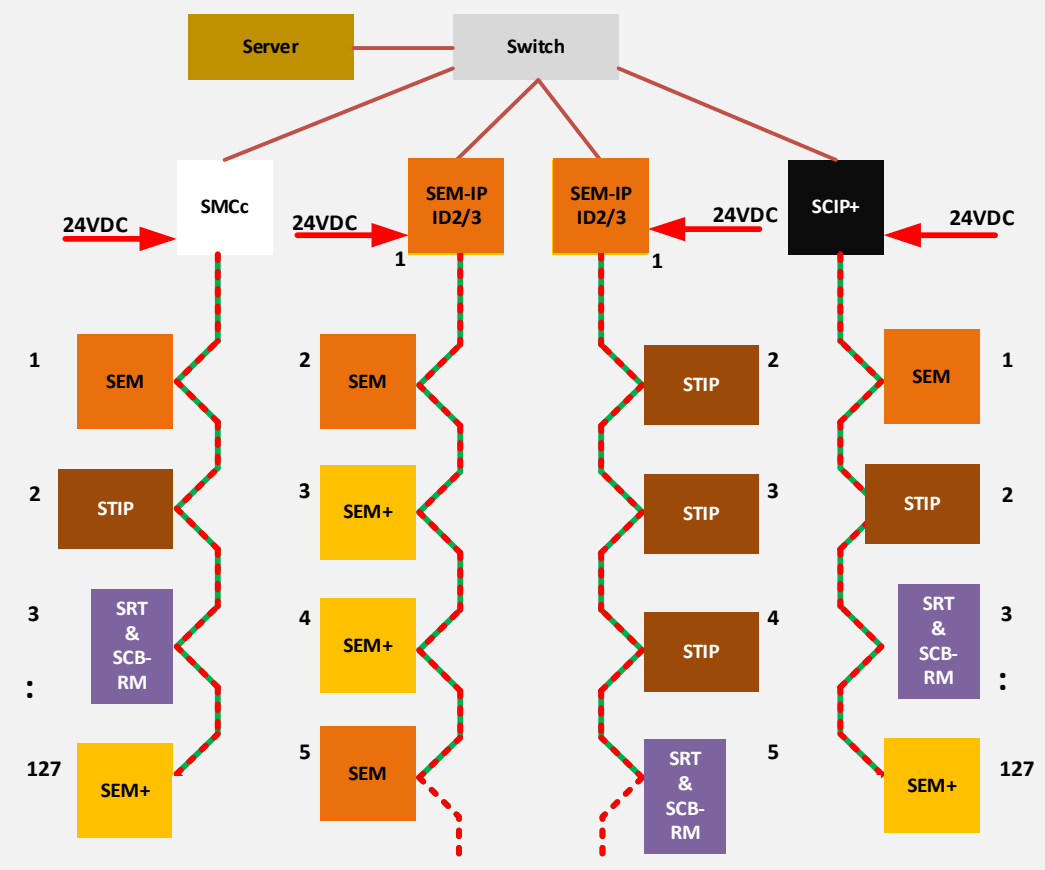
Bus start (factory setting)	Without termination	Bus end
Device terminates the start of ABUS and ZBUS	No termination ➔ device is on the bus in any position between the start and end device	Device terminates the end of ABUS and ZBUS
		

If changes are made to the termination, these must always be carried out in the same way on ABUS and ZBUS.

5.4 Connecting terminals and their function

Function	Connecting terminal	Properties
Power in 24 V / 10 A max.		**Mains power supply** from the ward, 24 V max. 10 A two 1.5 mm² wires possible per terminal.
2 X DBUS		**DBUS:** Use a twisted tone wire, max. length 400 m. 4 DBUS components possible per DBUS. Radial and mixed cabling possible (4 x 100 m).
ZBUS		**ZBUS:** Use a twisted tone wire, max. length 2500 m (without repeater max. 500 m). The ZBUS supports up to 127 ZBUS devices (use repeaters after 64 devices). Radial cabling is **not** permitted.
ABUS		**ABUS:** Use a twisted tone wire, max. length 2500 m (without repeater max. 500 m). The ABUS supports up to 127 ABUS devices (use repeaters after 64 devices). Radial cabling is **not** permitted.
RABUS		**R-ABUS: Room audio bus** Use a twisted tone wire, max. length 200 m. The R-ABUS supports up to 6 devices.
2 x NC (up) / passive I/O down		**I/O port:** Connection for passive call units exclusively in quiescent current. **I ➔** Call circuit input 0 V~24 V +/- 10% DC. **O ➔** Lamp output (reassurance light) max. 100 mA (open collector to GND)
2 x 24 V DC connection		**Mains power** for beds, max. 1.8 A (all mains power terminals for DBUS 1+2 together)

5.5 System overview ward controller & cluster controller

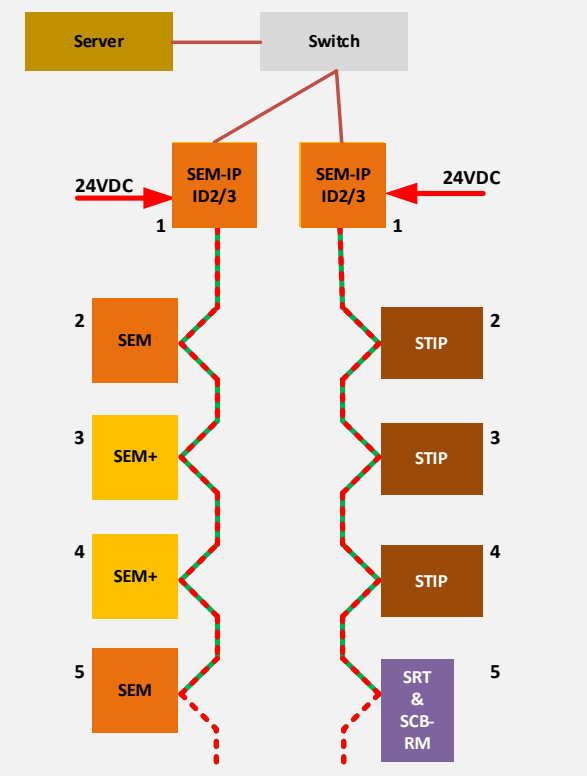


- An extension license is required in the case of over 5 rooms on the ZBUS
- 79SW421 for 5 additional rooms
- 79SW422 for 20 additional rooms

Legend	
Server	Systevo server
Switch	Network switch
SMCc	Systevo Control
SCIP+	Systevo Control IP+
SEM-IP ID2/3	Systevo Electronic Module IP SEM-IP (mode switch 2/3)
SEM	Systevo Electronic Module SEM
SEM+	Systevo Electronic Module plus SEM+
STIP	Systevo Touch Terminal STIP
SRT & SCB-RM	Systevo Room Terminal SRT
	LAN network connection
	System cable or comparable ZBUS/ABUS 24 V DC
	24 V DC external mains power supply. Ensure sufficient cable cross-section according to load.

Please check or inquire about the function availability.

5.6 System overview cluster controller



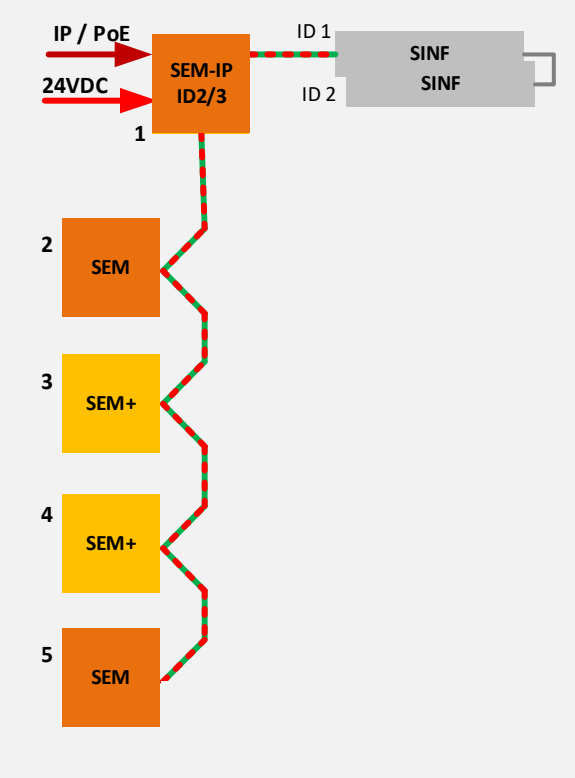
- An extension license is required in the case of over 5 rooms on the ZBUS
- 79SW421 for 5 additional rooms
- 79SW422 for 20 additional rooms

Legend	
Server	Systevo server
Switch	Network switch
SEM-IP ID2/3	Systevo Electronic Module IP SEM-IP (mode switch 2/3)
SEM	Systevo Electronic Module SEM
SEM+	Systevo Electronic Module plus SEM+
STIP	Systevo Touch Terminal STIP
SRT & SCB-RM	Systevo Room Terminal SRT
	LAN network connection
	System cable or comparable ZBUS/ABUS 24 V DC
	24 V DC external mains power supply. Ensure sufficient cable cross-section according to load.

Please check or inquire about the function availability.

5.7 System overview cluster controller with SINF connection to the ZBUS

System overview cluster controller with SINF on the ZBUS / ABUS



- An extension license is required in the case of over 5 rooms on the ZBUS
- 79SW421 for 5 additional rooms
- 79SW422 for 20 additional rooms
- For every SINF (single/double), one 79SW419 license is required

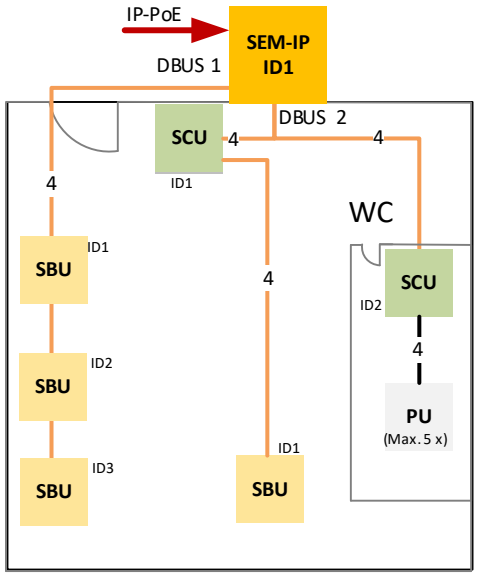
Legend	
SEM-IP ID2/3	Systevo Electronic Module IP (mode switch 2/3)
SINF	Systevo Information Display
SEM	Systevo Electronic Module
SEM+	Systevo Electronic Module plus
STIP	Systevo Touch Terminal
SRT & SCB-RM	Systevo Room Terminal
→	LAN network connection IP / PoE
	System cable or comparable ZBUS / ABUS 24 V DC
→	24 V DC external mains power supply. Ensure sufficient cable cross-section according to load.
	Included patch cable

Please check or inquire about the function availability.

5.8 IP-connected room (PoE)

5.8.1 Room without audio with 4 x SBU WC with 5 x passive buttons

IP-connected room (PoE) without audio



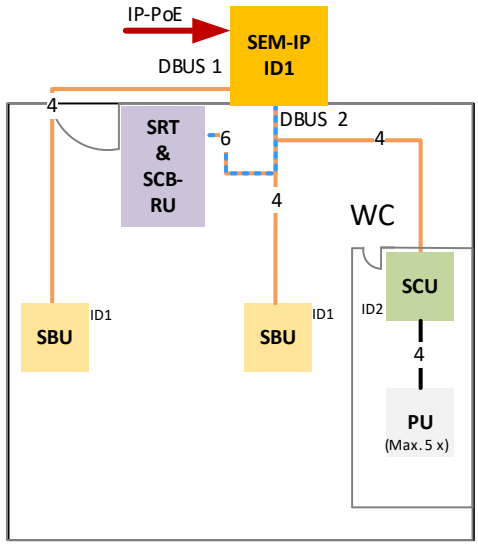
Only with Systevo Call Easy (max. 4).

Legend	
SEM-IP ID1	Systevo Electronic Module IP (mode switch 1)
SCU	Systevo Care Unit
SBU	Systevo Bed Unit
PU (Max. 5 x)	Passive call button / pull cord unit
→	LAN network connection IP / PoE

You must observe the maximum permitted power consumption of the components in the room (see also the table on output requirements in Chapter 4.2).

5.8.2 IP-connected room (PoE) with audio on the door

IP-connected one-bed room (PoE) with audio on the door



Only with Systevo Call Easy (max. 1).

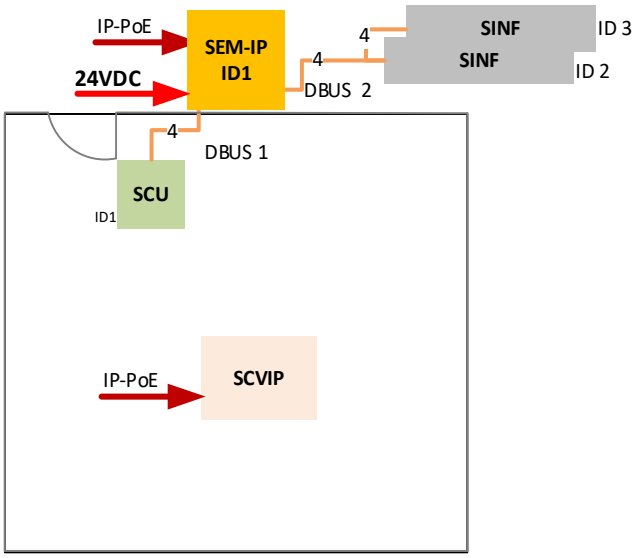
Legend	
SEM-IP ID1	Systevo Electronic Module IP (mode switch 1)
SCU	Systevo Care Unit
SBU	Systevo Bed Unit
SRT & SCB-RU	Systevo Room Terminal / Systevo Room Display with connection board RU
PU (Max. 5 x)	Passive call button / pull cord unit
→	LAN network connection IP / PoE
—4—	Passive tone wire 4 wires
—4—	DBUS 4 wires
-6-	DBUS and R-ABUS 6 wires

You must observe the maximum permitted power consumption of the components in the room (see also the table on output requirements in Chapter 4.2).

5.8.3 SEM-IP as a duty room with SINF connection to the DBUS

The SEM-IP is used as a corridor lamp and to connect a Systevo Care Unit. This ensures that audible call transfer via the Systevo Care Unit is guaranteed even when the Systevo Care View IP is unplugged. A direct connection of an information display SINF to the DBUS is possible in combination with the license (79SW419). The SINF is powered via the SEM-IP with an external 24 V DC mains power supply.

IP-connected duty room with SINF



- For operation of the Systevo Information Display (single/double) 79SW419 license is required
- Separate 24 V DC required for SINF
- Every SINF is connected to the DBUS
- A single SINF is connected with mode switch position 2
- Another SINF is connected with mode switch position 3
- A double-sided SINF is managed as two single SINFs
- A maximum of 2 SINFs can be connected

Legend	
SEM-IP ID1	Systevo Electronic Module IP (mode switch 1)
SCU	Systevo Care Unit
SINF	Systevo Information Display
SCVIP	Systevo Care View IP
→	LAN network connection IP / PoE
→	24 V DC external mains power supply for SINF
4	DBUS 4 wires

6 Commissioning

As standard on delivery (mode switch position 1), the SEM-IP works as an IP electronic module for rooms and is built into the system structure with an application on mobile end devices. This makes the process of locally assigning the installed devices much more convenient.

With the use of the mobile commissioning application, the previously standard initial commissioning process changes as follows:

1. **Preparation**
   a. In the CP99config configuration program, the required number of SEM-IP units to be installed are manually entered under the respective ward controller.
   b. At this point it is already possible to assign a room name to the components (SEM-IP). Alternatively, the name can be assigned at the location of the installed SEM-IP using the mobile application.
   c. Transfer the data to the mobile commissioning application (via IP network or USB cable of the mobile device).

2. **Creating a room list by scanning the QR code on the device using the mobile commissioning application.**
   a. After selecting the appropriate ward controller and the SEM-IP assigned to it, the QR code on the left side of the SEM-IP housing is read using a QR code scanner. The unique serial number of the device is linked to the room name.
   b. If a room name was already assigned to the SEM-IP during preparation, it can be used or adjusted as necessary.
   c. This process is repeated for all installed SEM-IP units.
   d. Once the device has been added to the room list, it is saved locally on the mobile device.
   e. Transfer the room list to the CP99config configuration program (via IP network or USB cable of the mobile device).

3. **Importing the data into the CP99config configuration program and transmitting the data to the system.**
   a. The previously created room list is transmitted from the mobile commissioning application to the CP99config configuration program.
   b. The read-in data is now adapted to the project requirements as usual with regard to the necessary nurse call parameters and settings.
   c. The finished configuration is written into the system.



- For clear identification, there is another QR code on the circuit board, which is preferably to be used.
- The data to be transferred (file extension .json) from cp99config is transferred manually via USB-C cable, e.g. to the download folder on the mobile device.
- A network connection from the mobile device to the server is required to use the mobile configuration. If there is no WiFi access to the server in the project, the following connections can be used:
  – USB-C ➔ LAN adapter
  – LAN ➔ WiFi adapter

7 Scope of delivery / accessories / licenses

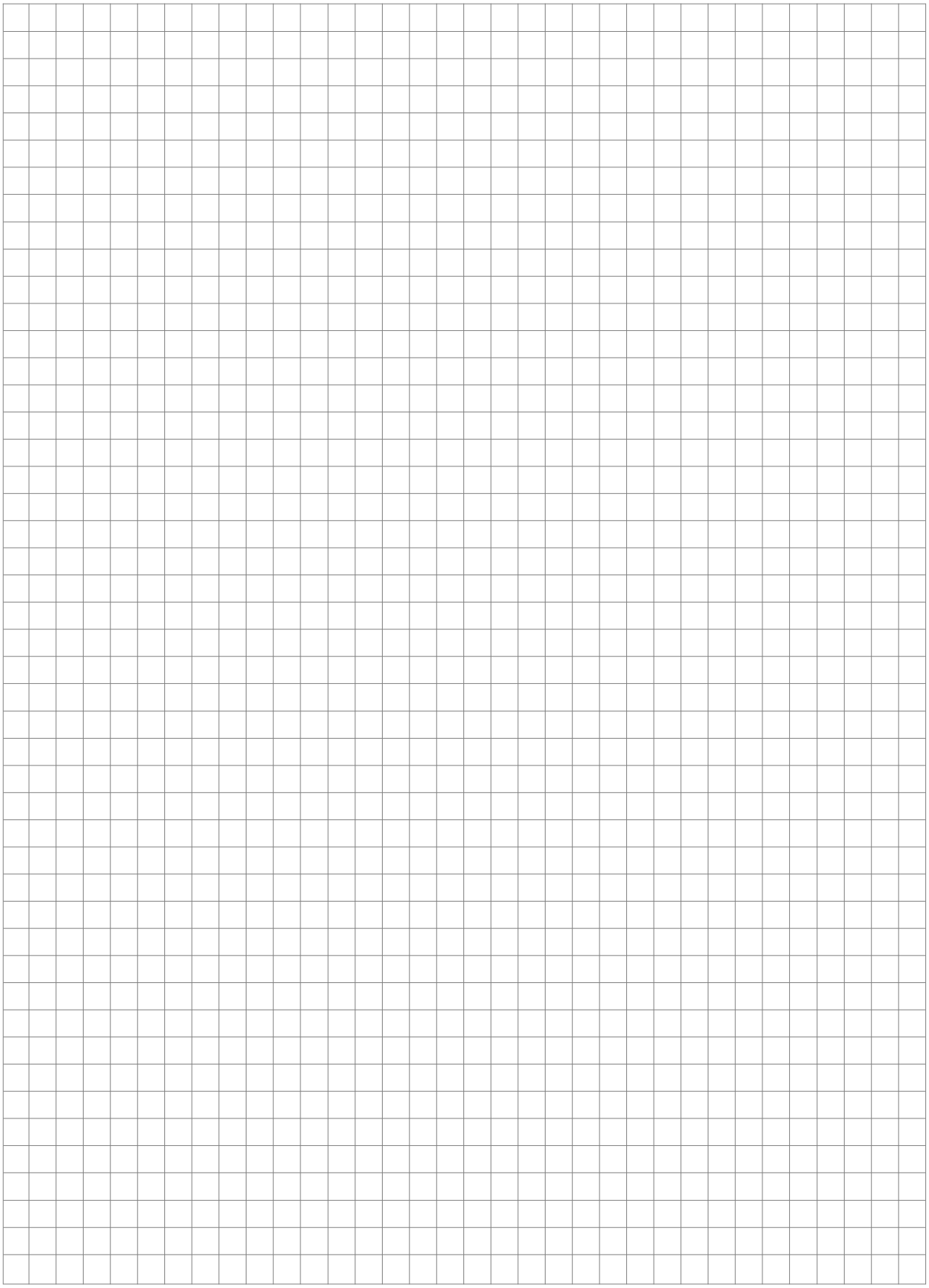
The Systevo Electronic Module is available as a bottom housing panel with the connection board in white and is equipped with covers in white or silver. For concealed installation whereby the lamps are not required, an opaque insert is available for the cover (it is already pre-assembled for part 72595L0). The name plate is also available in white and silver to match.

Part No.	Name
72595F	Systevo Electronic Module IP (SEM-IP)
72595L0	Dummy cover for 6 chambers
72595L1	Cover for 6 chambers (silver)
72595L2	Cover for 6 chambers (white)
72590T1	Name plate, silver
72590T2	Name plate, white
79CM307ZP	Set with connecting terminals for STIP / SEM / SEM+ / SINF / **SEM-IP**
Licences	
79SW419	Extension license for 1 Systevo Information Display on one Systevo Electronic Module IP
79SW420	Extension license for the basic activation of the “Cluster mode function”, including the first 5 rooms
79SW421	Extension license for the activation of the “Cluster mode function” (plus 5 rooms)
79SW422	Extension license for the activation of the “Cluster mode function” (plus 20 rooms)

8 Technical specifications

Systevo electronics module	SEM-IP (72595F)	Cover for SEM -IP (silver) (72595L1)	Cover for SEM-IP (white) (72595L2)	Cover insert for SEM-IP / (72595L0)
General				
Approvals/certifications	CE, VDE 0834, UL 1069 (component)			
Ambient conditions				
Application temperature	+5 °C – +40 °C			
Storage temperature	-20 °C – +70 °C			
Humidity	93% for 96 hours (non-condensing)			
Protection class	IP 20 (installed with attached lamp shell / cover)			
Electrical data				
Mains power supply	24 V DC +/- 10%, PoE	Supplied via SEM-IP		---
Current consumption in normal operation 2 LEDs light up white	145mA (an additional 28 mA for a name plate)	---		
ABUS termination	Bus, start termination, end termination	---		---
ZBUS termination	Bus, start termination, end termination	---		---
ZBUS topology / termination / length	Only BUS, slide switches on the circuit board 500 m	---		---
ABUS topology / termination / length	Only BUS, slide switches on the circuit board 500 m	---		---
R-ABUS topology / termination / length	Bus and radial, max. 200 m	---		---
DBUS ID / position	3 identical devices possible at the same DBUS line	---	---	---
Passive input	2 x 0 V – 24 V +/- 10%, NO/NC	---	---	---
Passive output	2 x max.125 mA open collector to GND	---	---	---
Connecting terminal supply	10 A, 1.5 mm²	---	---	---
Connecting terminals (data and signals)	Max. three 0.6 mm wires / two 0.8 mm wires per screw terminal	---	---	---
PoE type	Type 1	---	---	---
Max. output PoE / output for room components	13.2 W / 6.75 W	---	---	---
LAN protocol / network / speed / cable type	IEEE802.3af 10/100 Mbps Cable & patch cable Cat5 F/UTP (minimum), Length according to 802.3af	---	---	---
LAN network isolation	2xMOPP	---	---	---
USB	USB 2.0 OTG, USB-C connection	---	---	---
Mechanical specifications				
Dimensions of device (W x H x D)	110 x 133 x 19 mm Total dimensions in connection with L1 / L2 115 x 138 x 46 mm	115 x 138 x 33 mm		
Dimensions of the packaging (W x H x D)	172 x 57 x 124 mm			
Weight	248 g	172 g		157 g
Antimicrobial	Yes	No Lamp window = no	Yes Lamp window = no	Yes
UV resistant	Yes			
Fire protection class	V-0			
Housing material	PC + ABS-FR3010 (8% antimicrobial)	PC + ABS-FR3010 (8% antimicrobial) lamp window PC2407	PC + ABS-FR3010 (8% antimicrobial) lamp window PC2407	PC + ABS-FR3010 (8% antimicrobial)
Housing colour	White (similar to RAL 9003)	Silver	White (similar to RAL 9003)	White (similar to RAL 9003)
Wireless connections:				
WLAN	Not currently available	---	---	---
Bluetooth	Not currently available	---	---	---
Hygiene				
Disinfectants	70% isopropyl alcohol, Clorox bleach 10%, Descosept AF wipes, Sterillium (floor), Sani-Cloth HB, Sani-Cloth Plus, Super Sani-Cloth, 70% isopropyl alcohol wipes, Virex 256, CaviWipes, 409 glass and surface cleaner, Windex Blue, gentle dish soap and water (Dawn)			

Notes



Notes

Novar GmbH a Honeywell Company
Forumstraße 30
41468 Neuss, Germany
Internet: www.ackermann-clino.com
E-Mail: info@ackermann-clino.com

Technical changes reserved!
© 2024 Honeywell International Inc.